

Alison T. Cribb, PhD
Schmidt AI in Science Fellow, University of Oxford
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EDUCATION

- Ph.D. Geological Sciences, University of Southern California** 2018 - 2023
Los Angeles, California, USA
Doctoral dissertation title: “The geobiological role of bioturbating ecosystem engineers during key evolutionary intervals in Earth history”
Advisors: Prof. Frank A. Corsetti, Prof. David J. Bottjer
- B.A. Earth and Environmental Sciences, Vanderbilt University** 2014 - 2018
Conferred with Highest Honors
Nashville, Tennessee, USA
Honours thesis title: “Terminal Neoproterozoic bioturbation and implications for the extinction of the Ediacara biota”
Advisor: Dr. Simon A. F. Darroch

PROFESSIONAL APPOINTMENTS

- Schmidt AI in Science Fellow, University of Oxford** 2026 - 2028
Department of Earth Sciences, University of Oxford
Funded by an Eric and Windy Schmidt AI in Science Postdoctoral Fellowship, a programme of Schmidt Sciences
- 1851 Research Fellow, University of Southampton** 2023 - 2026
School of Ocean and Earth Science
Funded by a Science and Engineering Research Fellowship from the Royal Commission for the Exhibition of 1851

AWARDS, HONORS, AND FELLOWSHIPS

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| Eric and Wendy Schmidt AI in Science Postdoctoral Fellow | Schmidt Sciences; University of Oxford | 2026 - 2028 |
| Science and Engineering Research Fellowship | Royal Commission for the Exhibition of 1851 | 2023 - 2026 |
| PhD Achievement Award
<i>for academic excellence and success during doctoral studies, one of six awards across graduating PhDs</i> | University of Southern California | 2023 |
| Council Poster Prize
<i>for best poster among early career researchers</i> | Palaeontological Association | 2022 |
| Winifred Goldring Award, Honourable Mention | Association for Women Geoscientists; The Paleontological Society | 2022 |
| Gold Fellowship
<i>for research excellence among PhD candidates</i> | University of Southern California | 2021 |

President's Prize <i>for best talk among early career researchers</i>	Palaeontological Association	2020
Cisco Systems Graduate Fellowship <i>awarded to a selected cohort of women pursuing doctorate degrees in science and engineering</i>	Women in Science and Engineering (WiSE) - University of Southern California	2020
Student Presentation Award, Honourable Mention	Geological Society of America, Geobiology and Geomicrobiology Division	2019
Degree conferred with Highest Honors	Vanderbilt University	2018

RESEARCH GRANTS

Eric and Wendy Schmidt AI in Science Postdoctoral Fellowship	Schmidt Sciences, University of Oxford	2026-2028
Science and Engineering Research Fellowship	Royal Commission for the Exhibition of 1851	2023 - 2026
Gold Fellowship	University of Southern California	2021
Graduate Student Research Grant	Geological Society of American	2019
Student Research Grant	Society for Sedimentary Geology (SEPM) Foundation	2019
Harry B. Whittington Award	Paleontological Society	2017
Eugene Vaughan Assistantship in Geology	Vanderbilt University	2017

PUBLICATIONS

13. Darroch, S.A.F., Casey, M.M., Clapham, M.E., Contreras, D.L., Cover, Z., Craffey, M., **Cribb, A. T.**, Djunistee, I.A.P., Gearty, W., Gibson, B.<., Gotelli, N.J., Hayes, R., Hull, P.M., Ibarra, D.E., Lee, J., Looy, C.V., Muddiman, B., Shupinski, A.B., Sjosten, S., Smith, F.A., Stigall, A.L., Tome, C.P., Tingle, K., Turk, K.A., Villaseñor, A., Wagner, P.J., Wagner, J.D., Wang, S.C., Lyons, S.K. 2026. Ecosystem engineering, regime shifts and the rise of animals. *In press, Current Biology*.
12. **Cribb, A. T.**, Darroch, S.A.F., Gearty, W. Phanerozoic marine biodiversity increased in presence of ecosystem engineers. *Nature Communications*, 17, 9166. (doi: 10.1038/s41467-026-73628-6)

11. Feng, X., Chen, Z.Q., Benton, M., Bottjer, D.J., **Cribb, A. T.**, Zhao, L., Jiang, Y., Zhao, H., Yan, P., Huang, Y. Predator-prey interactions in the Early Triassic ocean. *GSA Bulletin*, 138, 3807-3818. (doi: 10.1130/B38121.1)
10. Darroch, S.A.F., Casey, M.M., **Cribb, A. T.**, Bates, A.E., Clapham, M.E., Contreras, D.L., Craffey, M., Duijnste, I.A.P., Gearty, W., Gotelli, N.J., Hamilton, M.J., Hayes, R.F., Hull, P.M., Ibarra, D.E., Lee, J., Looy, C.V., Lyson, T.R., Muddiman, B., Roopnarine, P.D., Shupinski, A.B., Smith, F.A., Stigall, A.L., Tome, C.P., Turk, K.A., Viallaseñor, A., Wagner, P.J., Wang, S., Lyons, S.K. 2025. Earth systems engineers and the lasting impact of organisms in deep time. *Trends in Ecology & Evolution*, 40, 1066-1076. (doi: 10.1016/j.tree.2025.08.005)
9. Tarhan, L.G., Pippenger, K., **Cribb, A. T.**, Zill, M., Phelps, W., Savrda, C.E., Droser, M.L., Bottjer, D.J., Clapham, M.E. 2025. Tracking bioturbating through time: The evolution of the marine sedimentary mixed and transition layers. *Science Advances*, 11, eadu7719. (doi: 10.1126/sciadv.adu7719)
8. **Cribb, A. T.**, Darroch, S.A.F. 2024. How to engineer a habitable planet: the rise of marine ecosystem engineers through the Phanerozoic. *Palaeontology*, 67, e12726. (doi: 10.1111/pala.12726) (invited contribution)
7. **Cribb, A. T.**, Formoso, K., Woolley, H. C., Beech, J., Brophy, S., Byrne, P. J., Cassady, V. C., Godbold, A. L., Larina, E., Maxeiner, P., Wu, Y.H., Corsetti, F. A., and Bottjer, D. J. 2023. Contrasting terrestrial and marine ecospace dynamics after the end-Triassic mass extinction event. *Proceedings of the Royal Society B*, 290, 20232232. (doi: 10.1098/rspb.2023.2232)
6. **Cribb, A. T.**, van de Velde, S. J., Berelson, W. M., Bottjer, D. J., and Corsetti, F. A. 2023. Ediacaran-Cambrian bioturbation did not extensively oxygenate sediments in shallow marine ecosystems. *Geobiology*, 24, 435-453. (doi: 10.1111/gbi.12550)
5. Feng, X., Chen, Z., Benton, M. J., Su, C., Bottjer, D.J., **Cribb, A. T.**, Li, Z., Zhao, L., Zhu, G., Huang, Y., and Guo, Z. 2022. Resilience of infaunal ecosystems during the Early Triassic greenhouse Earth. *Science Advances*, 8, eabo0596 (doi: 10.1126/sciadv.abo0597).
4. Darroch, S. A. F., **Cribb, A. T.**, Buatois, L. A., Germs, G. J. B., Kenchington, C. G., Smith, E. F., Mocke, H., O'Neil, G. F., Schiffbauer, J., Maloney, K., Racicot, R. A., Turk, K. A., Gibson, B. M., Koester, B. E., Boag, T. H., Tweedt, S., and Laflamme, M. 2020. The trace fossil record of the Nama Group, Namibia: Exploring the Ediacaran roots of the Cambrian Explosion. *Earth-Science Reviews*, 103435. (doi: 10.1016/j.earscirev.2020.103435)
3. Maloney, K. M., Boag, T. H., Faccioli, A. J., Gibson, B. M., **Cribb, A. T.**, Koester, B. E., Kenchington, C. G., Racicot, R. A., Darroch, S. A. F., and Laflamme, M. Paleoenvironmental analysis of Ernietta-bearing Ediacaran deposits in southern Namibia. 2020. *Palaeogeography, Palaeoclimatology, Palaeoecology*, 556, 109884. (doi:10.1016/j.palaeo.2020.109884)
2. **Cribb, A. T.**, Bottjer, D. J. 2020. Complex marine bioturbation ecosystem engineering behaviors persisted in the wake of the end-Permian mass extinction. *Scientific Reports*, 10, 203. (doi:10.1038/s41598-019-56740-0)

1. **Cribb, A. T.**, Kenchington, C. G., Koester, B. E., Gibson, B. G., Boag, T. H., Racicot, R. A., Mocke, H., Laflamme, M., and Darroch, S. A. F. 2019. Increase in metazoan ecosystem engineering prior to the Ediacaran-Cambrian boundary in the Nama Group, Namibia. *Royal Society Open Science*, 6, 190548. (doi:10.1098/rsos.190548)

Published perspective, news, and opinion pieces

14. **Cribb, A. T.** & Darroch, S.A.F. 2024. Machine learning maps Earth's hidden bioturbation. *Current Biology*, 34, 630-632. (doi: 10.1016/j.cub.2024.05.038)

In prep, submitted, and in review

Manuscripts in review or in revision

15. Wood, R.A., Bowyer, F., Anderson, R., **Cribb, A. T.**, Liu, A.G., Shields, G.A., Stockey, R.G., Erwin, D.H., Beyond a global narrative for the Ediacaran-Cambrian rise of animals. *In revision*.

Co-authored manuscripts in preparation

16. Djunistee, I.A.P., Hamilton, M.J., Darroch, S.A.F., Bates, A.E., Casey, M.M., Clapham, M.E., **Cribb, A. T.**, Contreras, D.L., Craffey, M., Gearty, W., Gotelli, N.J., Hayes, R.F., Hull, P.M., Ibarra, D.E., Lee, J., Looy, C.V., Lyson, T.R., Muddiman, B., Roopnarine, P.D., Shupinski, A.B., Smith, F.A., Stigall, A.L., Tome, C.P., Turk, K.A., Viallaseñor, A., Wagner, P.J., Wang, S., Lyons, S.K.. The paleobiology of human impacts on Earth systems. *In prep for PNAS*.
17. Stockey, R.G., **Cribb, A. T.**, Gruszka, P., Allen, B.J., Dean, C., Bowyer, F., Hulse, D., Pohl, A., Gearty, W., Monarrez, P. Overestimated Permian-Triassic marine extinction due to fossil record bias and heterogeneous ocean deoxygenation. *In prep for Nature Geoscience*.
18. Stockey, R.G., Li, N., Planavsky, N.J., Pohl, A., Finnegan, S., Sperling, E.A., **Cribb, A.T.**, Edwards, C., Dengh, Y., Goldberg, S., Saltzman, M.R., Dahl, T.W., Bergmann, K., Wei, G.Y., Kröger, B., Hopkins, M., Ridgwell, A., Fan, J., Shen, S., Zhang F. Cooling drove shallow ocean oxygenation and promoted the Great Ordovician Biodiversification Event while deep oceans remained anoxic. *In prep for Proceedings of the National Academy of Sciences*

Software development and contributions

19. Stockey, R.G., **Cribb, A. T.**, Gruszka, P., Sartin, A., Monarrez, P. PaleoClimR. R package. (in development: github.com/richardstockey/PaleoClimR)

INVITED PRESENTATIONS AND SEMINARS

European Geosciences Union – Invited speaker

2026

Session: Biogeodynamics and the Co-Evolution of Earth and Life
Vienna, Austria

University of Bristol, School of Earth Sciences/School of Biological Sciences

2025

Palaeontology seminar, Palaeobiology Research Group
Bristol, England, UK

Cambridge University, Department of Earth Sciences <i>Department of Earth Sciences seminar</i> Cambridge, England, UK	2025
Geological Society of America Annual Meeting – Keynote speaker <i>Session: Phylogenetic and Computational Approaches in Paleobiology</i> Anaheim, California, USA	2024
University of Oxford, All Souls College and Department of Earth Sciences <i>“The Past and Future of Life on our Dynamic Planet” seminar series</i> All Souls College, University of Oxford, England, UK	2024
University of Portsmouth, School of the Environment, Geography, and Geosciences <i>School of the Environment, Geography, and Geoscience seminar</i> Portsmouth, England, UK	2022
University of Leeds, School of Earth and Environment <i>Earth Surface Science Institute (ESSI) seminar series</i> Leeds, England, UK	2022
Biosphere Evolution, Transitions and Resilience (BETR) – Keynote speaker Milton Keynes, England, UK	2022
6th Annual Nereis Park Conference – Plenary lecture <i>Session: Bioturbation-Ecosystem Relationships</i> The European Institute for Marine Sciences (IUEM), Lagonna-Daoulas, Brittany, France	2022
Goldschmidt Annual Meeting – Keynote speaker <i>Session: Life and the Environment in the Phanerozoic</i> Honolulu, Hawaii, USA	2022
NASA Jet Propulsion Laboratory <i>Origins and Habitability Lab seminar</i> Pasadena, California, USA	2020

TEACHING

Student supervision and mentorship

Katie Mills <i>Predicting coral reef latitudinal biodiversity gradients under various SSP scenarios using novel agent-based models</i>	MSci Marine Biology with Oceanography, University of Southampton	2025 - 2026
Bel Holmes <i>Coral extinction and recovery patterns at the Triassic-Jurassic boundary</i>	BSci Marine Biology, University of Southampton	2024 - 2025
Neha Ghosh <i>Terrestrial ecological responses to ancient and modern extinctions</i>	MSci Natural Sciences (third-year), University of Southampton	2024 - 2025

Neha Ghosh

A Tale of Two Warming Events: Comparing ecological responses of modern and ancient climatic extremes

SMMI Undergraduate
Summer Intern, University of
Southampton 2024

Instructor

Earth and Ocean System	University of Southampton	2023, 2024
Palaeobiology	University of Southampton	2024
Marine Invertebrates	University of Southampton	2024
Co-evolution of Life and Earth	University of Southampton	2024

Teaching assistant

The History of Life on Earth: A View through the Museum	University of Southern California	2019, 2020, 2021
Climate Change	University of Southern California	2019
Crises of a Planet	University of Southern California	2018
Earth Systems Through Time	Vanderbilt University	2016, 2017

Pedagogical training

Practicum in Teaching the Liberal Arts <i>Focus on teaching in the geological and biological sciences</i>	University of Southern California	2018
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PROFESSIONAL SERVICE

Committees, appointments, and research networks

Vice-chair of the Earth System Science Group	Geological Society of London	2026-present
Secretary of the Earth System Science Group	Geological Society of London	2023-2026
EDI Chair of the Earth System Science Group	Geological Society of London	2025-present
Ecological and Evolutionary Effects of Extinction and Ecosystem Engineers (E6) RCN	National Science Foundation-funded Research Coordination Network	2021-present
Graduate Student Representative	Geological Society of America – Geobiology and Geomicrobiology division	2020-2022
Dean's Student Advisory Board	Vanderbilt University	2016-2018

Academic meetings and seminars

Life and Planet 2023 <i>Organizing committee member</i>	Geological Society of London	2026-present
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International Fossil Coral and Reef Society Early Career Research Symposium <i>Lead organizing committee member</i>	International Fossil and Coral Reef Society	2026
International Fossil Coral and Reef Society Early Career Research Symposium <i>Deputy organizing committee member</i>	International Fossil and Coral Reef Society	2025
Symposium of Vertebrate Palaeontology and Comparative Anatomy <i>Organizing committee member</i>	University of Southampton	2024
Paleo/Environment Seminar <i>Co-organiser of weekly seminars for the academic year</i>	University of Southern California	2021-2022
Southern California Geobiology Symposium <i>Deputy co-chair of organizing committee</i>	University of Southern California (ran virtually)	2020-2021
<u>Conference sessions</u>		
Ecosystem Engineering Through Earth History <i>Co-chaired with other graduate student members of the E6 RCN</i>	Geological Society of America Denver, Colorado	2022
New Voices in Geobiology <i>Annual GSA GBGM Division Session</i>	Geological Society of America Denver, Colorado	2022
New Advances in Geobiology <i>Annual GSA GBGM Division Session</i>	Geological Society of America Denver, Colorado	2022
New Voices in Geobiology <i>Annual GSA GBGM Division Session</i>	Geological Society of America Portland, Oregon	2021
New Advances in Geobiology <i>Annual GSA GBGM Division Session</i>	Geological Society of America Portland, Oregon	2021
Ecosystem Engineering Through Earth History	Geological Society of America Virtual Meeting	2020
New Voices in Geobiology <i>Annual GSA GBGM Division Session</i>	Geological Society of America Virtual Meeting	2020
New Advances in Geobiology <i>Annual GSA GBGM Division Session</i>	Geological Society of America Virtual Meeting	2020
Phanerozoic Earth-Life Transitions <i>Invited student session chair</i>	Geobiology Society Conference Banff, Canada	2019

Other service roles

President's Prize judge <i>Judge for PalAss President's Prize for best three student talks at annual meeting</i>	The Palaeontological Association	2024
Student presentation judge <i>GSA Geobiology and Geomicrobiology Division student presentation awards at the GSA Annual meeting</i>	Geological Society of America – Geomicrobiology and Geobiology Division	2023, 2024
Student presentation judge <i>Paleontological Society student presentation awards at the GSA Annual Meeting</i>	The Paleontological Society	2023, 2024

Reviewer

Journals: *Current Biology*; *Geobiology*; *Geological Magazine*; *Journal of Open Source Science*; *Nature Communications Biology*; *Paleobiology*; *Palaeogeography*, *Palaeoclimatology*, and *Palaeoecology*; *Palaeontology*; *Proceedings of the National Academy of Sciences*; *Proceedings of the Royal Society B*; *Scientific Reports*; *Science Advances*

Funding agencies: *American Chemical Society*

Professional membership

Fellow of the Geological Society (FGS); Geological Society of London Earth System Science Group; the Palaeontological Association

PUBLIC OUTREACH AND ENGAGEMENT

Southampton Science and Engineering Festival (SOTSEF) <i>Activity leader: "Ocean Time Machine: Explore reef fossils and ancient climates"</i>	2025
Winchester Cathedral Whales <i>Design of evolution of whales and marine megafauna exhibit</i>	2025
Sidmouth Science Festival <i>Volunteer with University of Southampton, School of Ocean and Earth Science Developed and presented exhibit "Reefs Through Time"</i>	2024
Southampton International Boat Show <i>Volunteer Exhibition of "Reefs Through Time" in University of Southampton tent</i>	2024
Lyme Regis Fossil Festival <i>Organizing team and volunteer with University of Southampton, School of Ocean and Earth Science Developed and presented exhibit "Reefs Through Time"</i>	2024
Southampton Science and Engineering Festival (SOTSEF) <i>School of Ocean and Earth Science organizing team Assisted in development of exhibit on coral reefs, mammoths, and ichthyosaurs</i>	2024
Stem Bytes Seminar Speaker, Women in Science and Engineering (WiSE) chapter of the University of Southern California <i>"How Early Worms Helped Engineer Earth's Early Benthic Ecosystems" Public talk aimed to increase recruitment of women into STEM majors at the University of Southern California.</i>	2020

On Point, National Public Radio (NPR) guest

2018

“It’s Senior Thesis Season” episode

Invited to discuss undergraduate honors thesis to a national audience on live radio

CONFERENCE ABSTRACTS

Abridged: First-author abstracts presented since 2020

[†]Invited presentation

[§]Award given for presentation

- [†]24. Cribb, A.T., Sartin, A., Allen, B., Stockey, R., Monarrez, P., Hulse, D. 2026. Ecological and biogeochemical consequences of benthic ecosystem engineer responses to the end-Permian mass extinction. *EGU General Assembly*. Vienna, Austria. Invited.
- 23. Cribb, A.T. 2025. What controlled the appearance of coral reef biodiversity hotspots throughout the Phanerozoic? *International Fossil Coral and Reef Society Early Career Symposium*. Online. Oral presentation.
- 22. Cribb, A.T. 2025. Responses and effects of marine ecosystem engineers during the end-Permian mass extinction. *Crossing the Paleontological-Ecological Gap & Conservation Paleobiology Symposium*, Zurich, Switzerland. Oral presentation.
- 21. Cribb, A.T. 2025. Ecological and climatic controls on coral reef biodiversity hotspots through the Phanerozoic. *Life and Planet*, London, England. Oral presentation. Highlight talk.
- 20. Cribb, A.T. 2024. Do ecosystem engineers facilitate recovery after mass extinctions? *Palaeontological Association Annual Meeting*. Erlangen, Germany. Poster presentation.
- [†]19. Cribb, A.T. 2024. Role of ecosystem engineers in Phanerozoic biodiversity and ecology. *Geological Society of America Annual Meeting*, Anaheim, CA. Oral presentation. Invited keynote for “Phylogenetic and Computational Approaches in Paleobiology.”
- 18. Cribb, A.T., Darroch, S.A.F., Gearty, W., Ezard, T.H.G. 2024. Phanerozoic marine biodiversity increased in presence of ecosystem engineers. *Life and Planet*, London, England. Poster presentation.
- 17. Cribb, A.T., Darroch, S.A.F., Ezard, T.H.G. 2024. Ecosystem engineers impact marine biodiversity through the Phanerozoic. *European Geosciences Union*, Vienna, Austria. Oral presentation (virtual).
- 16. Cribb, A.T., Stockey, R.G., Feng, X., Bottjer, D.J. 2023. Environmental drivers and biogeochemical consequences of benthic extinctions in the Early Triassic. *Geological Society of America Annual Meeting*, Pittsburgh, PA, USA. Oral presentation.
- 15. Cribb, A.T., Darroch, S.A.F., Ezard, T.H.G. 2023. How to engineer a habitable planet: New frameworks for understanding the rise and fall of marine ecosystem engineers through Earth history. *The Palaeontological Association Annual Meeting*, Cambridge, England. Oral presentation. Selected for the annual meeting symposium, “Ecosystem engineers through deep time”

14. Cribb, A.T., Stockey, R.G., Feng, X., Bottjer, D.J. 2023. Biogeochemical consequences of benthic extinctions during the end-Permian mass extinction. *Life and Planet*, London, England. Poster presentation.
13. Cribb, A.T., Formoso, K., Woolley, H.C., Beech, J., Brophy, S., Byrne, P.J., Cassady, V.C., Godbold, A.L., Larina, E., Maxeiner, P., Wu, Y.H., Corsetti, F.A., Bottjer, D.J. 2022. Decoupled terrestrial and marine ecological recovery after the end-Triassic mass extinction event. *Geological Society of America Annual Meeting*. Denver, CO. Hybrid oral presentation.
- [†]12. Cribb, A.T., van de Velde, S.J., Berelson, W.M., Corsetti, F.A., Bottjer, D.J. 2022. Investigating macrofauna-driven sediment oxygenation during the Ediacaran-Cambrian transition. *Biosphere, Evolutions, and Transitions (BETR) finale meeting*. Milton-Keynes, England. Invited oral presentation.
- [†]11. Cribb, A.T. Utilization of the trace fossil record to understand bioturbators' ecosystem engineering impact over the last 560 million years. *6th Nereis Park Conference and Thematic School*, Lagonna-Daoulas, France. Invited plenary speaker for session "Bioturbation-ecosystem relationships."
- [§]10. Cribb, A.T., Godbold, A.L., van de Velde, S.J., Koester, B.E., Celestian, A., Bottjer, D.J., Corsetti, F.A. 2022. Bioturbating ecosystem engineers and their sedimentary biogeochemical impact: A case study from the Lower Cambrian Deep Spring Formation. *The Palaeontological Association Annual Meeting*, Cork, Ireland. Poster presentation. Awarded Annual Meeting Council Poster Prize.
9. Cribb, A.T., Formoso, K., Woolley, C.H., Beech, J.D., Brophy, S., Byrne, P.J., Cassady, V.C., Godbold, A.L., Larina, E., Maxeiner, P., Wu, Y., Corsetti, F.A., Bottjer, D.J. 2022. Terrestrial and marine ecospace dynamics across the end-Triassic mass extinction event. *The Palaeontological Association Annual Meeting*, Cork, Ireland. Poster presentation.
- [†]8. Cribb, A.T., van de Velde, S.J., Berelson, W.M., Bottjer, D.J., Corsetti, F.A. 2022. Geobiological impacts of Earth's early complex bioturbators during the Ediacaran-Cambrian transition. *Goldschmidt Annual Meeting*, Hawaii, USA. Invited keynote speaker in session "Life and the Environment in the Phanerozoic."
7. Cribb, A.T., van de Velde, S.J., Berelson, W. M., Corsetti, F. A., Bottjer, D. J. 2022. Revealing the biogeochemical role of Ediacaran-Cambrian bioturbators during the Agronomic Revolution. *Southern California Geobiology Symposium*, Riverside, CA, USA. Oral presentation.
- [§]6. Cribb, A.T., van de Velde, S.J., Berelson, W.M., Corsetti, F.A., Bottjer, D.J. 2021. Did early bioturbators oxygenate shallow to deep sediment tiers in Ediacaran-Cambrian benthic ecosystems? *Geological Society of America Annual Meeting*, Portland, OR, USA. Oral presentation. Selected for The Paleontological Society's session Future Leaders in Paleontology.
5. Cribb, A.T., Bottjer, D.J. 2021. Biogeochemical consequences of a sedimentary mixed layer collapse in the wake of the end-Permian mass extinction. *Geological Society of America Annual Meeting*, Portland, OR, USA. Poster presentation.
4. Cribb, A.T., van de Velde, S.J., Berelson, W.M., Bottjer, D.J., Corsetti, F.A. 2021. Controls on the oxygen penetration depth in Ediacaran-Paleozoic benthic ecosystems: A reactive-transport modeling study. *Goldschmidt Annual Meeting*, online meeting. Online oral presentation.

- [§]3. Cribb, A.T., van de Velde, S.J., Darroch, S.A.F., Bottjer, D.J., Corsetti, F.A. 2020. Growing pains of the Agronomic Revolution: Early bioturbators stimulated sulfide production in shallow sediment tiers. *The Palaeontological Association Annual Meeting*, online meeting. Online oral presentation. Awarded the President's Prize.
2. Cribb, A.T., van de Velde, S.J., Darroch, S.A.F., Bottjer, D.J., Corsetti, F.A. 2020. Engineering benthic biogeochemistry: Reactive-transport modeling of biomixing and bioirrigation behaviors across the Ediacaran-Cambrian boundary. *Geological Society of America Meeting*, GSA Connects Online. Online oral presentation.
1. Cribb, A.T. & Bottjer, D.J. 2020. Complex benthic ecosystem engineering in the Early Triassic: The trace fossil record as a tool for understanding ecosystem function after mass extinction events. *Ocean Sciences Meeting*, San Diego, CA. Poster presentation.